

ALEXANDER JAMES THORNTON

Senior Principal Engineer | Cloud Architect | AI/ML Platform Lead

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PROFESSIONAL SUMMARY

Results-driven **Senior Principal Engineer** with **14+ years** of progressive experience designing and delivering enterprise-scale cloud platforms, distributed systems, and AI/ML infrastructure across FinTech, HealthTech, and SaaS verticals. Proven record of leading cross-functional teams of 20–50 engineers, reducing operational costs by \$4M+, and shipping zero-downtime migrations on AWS, GCP, and Azure. Deep expertise in **Python, Java, Kotlin, Go, Terraform, Kubernetes, Apache Spark, and Apache Kafka**. Passionate about Engineering Excellence, Developer Experience (DevEx), and building high-trust remote-first teams.

Reduced cloud infrastructure costs by 38% (\$4.2M annually) through FinOps practices at MegaCorp Financial

Architected real-time data platform processing 2.4 billion events/day with 99.999% SLA using Apache Kafka & Flink

Led development of internal ML platform (MLOps) serving 180+ models in production, cutting model deployment time from 3 weeks to 4 hours

CORE COMPETENCIES & TECHNICAL SKILLS

Python / Django / FastAPI ★★★★★	Java / Spring Boot / Micronaut ★★★★★	Go / gRPC / Protobuf ★★★★★
Kotlin / Ktor / Coroutines ★★★★★	JavaScript / TypeScript / Node.js ★★★★★	Scala / Akka / Play ★★★★★
AWS (14 certifications) ★★★★★	GCP / BigQuery / Vertex AI ★★★★★	Azure / AKS / DevOps ★★★★★
Kubernetes / Helm / ArgoCD ★★★★★	Terraform / Pulumi / CDK ★★★★★	Docker / Podman / containerd ★★★★★
Apache Kafka / Confluent ★★★★★	Apache Spark / Flink / Beam ★★★★★	Apache Airflow / Dagster / Prefect ★★★★★
PostgreSQL / MySQL / Aurora ★★★★★	MongoDB / DynamoDB / Cassandra ★★★★★	Redis / Elasticsearch / OpenSearch ★★★★★
TensorFlow / PyTorch / JAX ★★★★★	MLflow / Kubeflow / SageMaker ★★★★★	Hugging Face / LangChain / RAG ★★★★★
GitHub Actions / Jenkins / CircleCI ★★★★★	Prometheus / Grafana / Datadog ★★★★★	OpenTelemetry / Jaeger / Zipkin ★★★★★

PROFESSIONAL EXPERIENCE

Senior Principal Engineer – Cloud Platform | MegaCorp Financial Technologies

Jan 2020 – Present (4 yrs 6 mos)
San Francisco, CA (Hybrid)

MegaCorp Financial Technologies is a \$12B fintech unicorn processing \$340B in annual payment volume across 42 countries.

- **Cloud Infrastructure Strategy:** Architected and owned the
- Designed multi-region active-active deployment on AWS spanning us-east-1, eu-west-1, and ap-southeast-1, achieving 99.999% uptime SLA for core payment APIs.
- Introduced Kubernetes-native service mesh (Istio + Envoy) replacing legacy HAProxy cluster, reducing p99 API latency from 420ms to 38ms.
- Led FinOps initiative implementing rightsizing, Reserved Instances, and Spot Fleet strategies — saving \$4.2M/year in cloud spend (38% reduction).
- Built internal Developer Platform (IDP) using Backstage, Crossplane, and ArgoCD enabling 240 engineers to self-serve infrastructure via GitOps.
- Drove Security & Compliance: achieved SOC 2 Type II, PCI DSS Level 1, ISO 27001 certifications by designing automated policy-as-code using OPA/Gatekeeper.
- Mentored 8 Senior Engineers toward Staff/Principal promotions; conducted 150+ technical interviews.
- Used Python, Go, Terraform, Kubernetes, AWS, Istio, ArgoCD, Backstage, OPA, PostgreSQL, Redis in this role.

Principal Engineer – Data & AI Platform | Streamline Analytics Inc.

Mar 2017 – Dec 2019 (2 yrs 9 mos)
Seattle, WA (On-site)

Streamline Analytics (acquired by Salesforce, 2021) – B2B SaaS analytics platform serving 3,000+ enterprise customers.

- Built real-time streaming data platform processing 2.4B events/day using Apache Kafka (300-node cluster), Flink, and Delta Lake on Databricks.
- Reduced end-to-end data pipeline latency from 45 minutes to under 90 seconds for financial reporting dashboards.
- Designed feature store (using Feast + Redis) serving 180 production ML models with sub-10ms feature retrieval at 50K RPS.
- Deployed self-service ML platform on Kubernetes (Kubeflow + MLflow), cutting model deployment cycle from 3 weeks to 4 hours.
- Partnered with Product to build customer churn prediction model (XGBoost) that improved retention by 14% (\$22M ARR impact).
- Contributed to open-source Kafka ecosystem; merged 17 PRs to Apache Kafka project (connector improvements and performance patches).
- Stack used: Python, Scala, Apache Kafka, Flink, Delta Lake, Databricks, Feast, Redis, Kubernetes, Kubeflow, MLflow, Terraform, GCP.

PROFESSIONAL EXPERIENCE (CONTINUED)

Lead Backend Engineer | HealthBridge Systems

Jun 2015 – Feb 2017 (1 yr 8 mos)
Austin, TX (On-site)

HealthBridge – HIPAA-compliant EHR + telehealth platform serving 400+ hospital networks and 2M+ patients.

- Led migration from monolithic Rails application to Java/Spring Boot microservices (32 services), enabling independent deployments and 10× throughput.
- Designed FHIR R4-compliant API gateway handling 8M+ API calls/day with zero HIPAA data breaches across 2+ years.
- Implemented OAuth 2.0 / SMART-on-FHIR authorization framework for EHR integrations with Epic, Cerner, and Allscripts.
- Built real-time clinical alerting engine using AWS Kinesis + Lambda processing 500K vital sign readings/minute.
- Reduced patient appointment scheduling time by 60% through AI-powered scheduling optimizer (OR-Tools + custom heuristics).
- Technologies: Java 11, Spring Boot, PostgreSQL, MySQL, AWS (EC2, RDS, Kinesis, Lambda, S3), Docker, Jenkins, FHIR R4.

Senior Software Engineer | NovaSoft Technologies

Aug 2013 – May 2015 (1 yr 9 mos)
Chicago, IL (On-site)

NovaSoft – SaaS HR & payroll platform serving 15,000+ SMB customers across North America.

- Re-architected payroll calculation engine in Python/Celery, processing 1.2M payroll runs/month with 4-nines reliability.
- Implemented automated tax compliance module integrating with IRS, SSA, and 50-state tax APIs — eliminating \$800K/year in manual auditing costs.
- Led frontend rebuild from jQuery to React (v0.14), improving page load times by 72% and boosting NPS by 18 points.
- Built CI/CD pipeline (Jenkins + Docker) reducing release cycle from bi-weekly to daily deployments.
- Stack: Python, Django, Celery, PostgreSQL, React, Redis, AWS EC2/RDS, Jenkins, Nginx.

Software Engineer | DataBridge Consulting Group

Jul 2011 – Jul 2013 (2 yrs)
New York, NY (On-site)

DataBridge – Boutique technology consulting firm specializing in financial data systems and ETL pipelines.

- Designed ETL pipelines for 12 enterprise clients (Goldman Sachs, JPMorgan, BNY Mellon) ingesting structured/unstructured financial data.
- Built market risk aggregation system in Java processing real-time FX rates, equities, and derivatives positions across 18 asset classes.
- Developed custom data quality framework validating 200+ business rules with automatic anomaly alerting via email/Slack.
- Delivered 6 client projects on-time and under budget across 2 years; achieved 98% client satisfaction score.
- Stack: Java, Oracle DB, Informatica, IBM DataStage, Python scripting, Microsoft SQL Server, UNIX shell scripting.

EDUCATION

M.S. Computer Science – Distributed Systems & Machine Learning | Stanford University

Sep 2009 – Jun 2011
Stanford, CA

- GPA: 3.92/4.0 | Thesis: 'Fault-Tolerant Consensus Protocols in Byzantine Distributed Systems'
- Teaching Assistant: CS246 – Mining Massive Datasets (Prof. Jure Leskovec)
- Relevant Coursework: Advanced Algorithms, Distributed Systems, Machine Learning, Compilers, Database Systems

B.S. Computer Engineering – Summa Cum Laude | University of Michigan, Ann Arbor

Sep 2005 – May 2009
Ann Arbor, MI

- GPA: 3.97/4.0 | Valedictorian, College of Engineering | Phi Beta Kappa Honor Society
- Senior Capstone: Distributed RAID storage system achieving 40% higher throughput than baseline (Ranked #1 in cohort)
- IEEE Student Chapter President (2008–2009) | ACM Programming Competition – 1st Place (2008)

CERTIFICATIONS & CREDENTIALS

AWS Certified Solutions Architect – Professional	Amazon Web Services	2023 (SAP-C02)
Google Cloud Professional Data Engineer	Google Cloud	2022
Certified Kubernetes Administrator (CKA)	CNCF	2023
Certified Kubernetes Security Specialist (CKS)	CNCF	2023
AWS Certified DevOps Engineer – Professional	Amazon Web Services	2022 (DOP-C02)
HashiCorp Certified: Terraform Associate	HashiCorp	2023
Google Cloud Professional Cloud Architect	Google Cloud	2021
Databricks Certified Machine Learning Professional	Databricks	2022
Confluent Certified Developer for Apache Kafka	Confluent	2021
MongoDB Certified Developer Associate	MongoDB University	2020

KEY PROJECTS & TECHNICAL ACHIEVEMENTS

PROJECT 1: Global Payment Orchestration Platform (MegaCorp)

Role: Technical Lead | Team: 22 Engineers | Duration: 18 Months | Budget: \$8.5M

- Designed event-sourced, CQRS-based payment orchestration system handling 45,000 TPS at peak with sub-50ms P99.
- Integrated 28 payment providers (Stripe, Adyen, Braintree, local APAC/LATAM providers) via a unified connector abstraction layer.
- Implemented saga pattern for distributed transactions ensuring exactly-once payment guarantees across microservices.
- Reduced payment failure rate from 2.1% to 0.08% through intelligent retry logic, circuit breakers, and provider health monitoring.

Tech: Python, Go, Kafka, Redis, PostgreSQL, AWS (EKS, RDS Aurora, ElastiCache, SQS), Terraform, Istio, Prometheus, Grafana

PROJECT 2: Enterprise ML Platform – 'Nexus AI' (Streamline Analytics)

Role: Platform Architect | Team: 14 Engineers + 6 Data Scientists | Duration: 12 Months

- Built end-to-end ML platform supporting data ingestion, feature engineering, model training, experiment tracking, model serving, and monitoring.
- Designed auto-scaling model serving infrastructure on Kubernetes supporting 2,200 inference RPS per model with GPU autoscaling.
- Implemented A/B testing framework enabling safe progressive model rollouts with automated statistical significance testing.
- Created metadata cataloging system tracking 14,000+ datasets, 3,200+ experiments, and 180+ production models.

Tech: Python, Kubeflow, MLflow, Feast, Redis, Spark, Delta Lake, Kubernetes, GKE, Vertex AI, PostgreSQL, Grafana

PROJECT 3: Developer Internal Platform – 'Forge' (MegaCorp)

Role: Principal Architect | Team: 8 Engineers | Duration: 9 Months | Users: 240 Engineers

- Built 'Forge', a golden-path developer platform reducing new service time-to-production from 3 weeks to 2 hours.
- Integrated Backstage (service catalog), Crossplane (cloud resources), ArgoCD (GitOps), and Vault (secrets management).
- Created 22 reusable Terraform module templates covering compute, data, networking, security, and observability stacks.
- Developer satisfaction score (DX survey) improved from 54% to 91% post-launch; platform adopted by 100% of engineering org within 3 months.

Tech: TypeScript, Go, Backstage, Crossplane, ArgoCD, HashiCorp Vault, Terraform, Kubernetes, GitHub Actions, Datadog

ADDITIONAL PROJECTS

PROJECT 4: Zero-Downtime Database Migration (HealthBridge)

- Migrated 2.4TB production PostgreSQL database from on-prem to AWS Aurora with zero downtime using dual-write + shadow-read pattern.
- Achieved 4× query performance improvement through schema redesign, index optimization, and query plan analysis.
- Wrote custom replication validation tool (12K lines Python) verifying 100% data fidelity across 180+ tables.

PROJECT 5: Real-Time Fraud Detection System (MegaCorp)

- Built ensemble fraud detection system (XGBoost + neural network) scoring every transaction in <12ms at 15K TPS.
- Reduced false positive rate by 62% vs. rule-based system while maintaining 98.7% fraud recall.
- Prevented \$78M in fraudulent transactions in first year; saved \$4.2M in chargeback fees.

PROJECT 6: Open Source – KafkaLens Monitoring Dashboard

- Created open-source Kafka cluster monitoring dashboard (React + Go) — 4,200+ GitHub stars, 180+ contributors, downloaded 22K+ times.
- Integrated with Prometheus, Grafana, and Datadog for seamless enterprise observability workflows.

LEADERSHIP & MANAGEMENT

- Managed and mentored distributed engineering teams of up to 22 engineers across 4 time zones (US, India, UK, Poland).
- Conducted 150+ technical interviews; calibrated engineering ladders from L3 to Staff/Principal levels.
- Facilitated quarterly architecture review board (ARB) meetings driving org-wide alignment on technical direction.
- Established Engineering Excellence Program: blameless postmortems, error budget policies, and SLO frameworks for 40+ services.
- Championed Diversity & Inclusion: grew women-in-engineering ratio on my teams from 18% to 36% over 3 years.
- Delivered 20+ tech talks at internal All-Hands, Brown Bags, and external conferences (KubeCon, AWS re:Invent, PyCon).
- Authored Engineering Wiki (120+ pages) adopted as onboarding standard for entire 240-person engineering org.
- Budget ownership: managed \$12M annual cloud + tooling budget; consistently delivered 5–15% under-budget.

PUBLICATIONS & RESEARCH

Conference Papers

- Thornton, A.J. et al. (2023). 'FinOps at Scale: Automated Cloud Cost Optimization Patterns for Financial Services.' IEEE CloudCom 2023, Naples, Italy.
- Thornton, A.J. & Rodriguez, M. (2022). 'Stateful Stream Processing Reliability: Lessons from 2.4B Events/Day.' VLDB 2022, Sydney, Australia.
- Thornton, A.J. (2021). 'Building Developer Platforms that Actually Get Adopted: Patterns and Anti-Patterns.' PlatformCon 2021 (online).
- Thornton, A.J. (2020). 'Real-Time Fraud Detection at Scale: Hybrid ML Architectures for Financial Transactions.' ACM KDD 2020, San Diego, CA.
- Thornton, A.J. & Liu, W. (2019). 'Feature Store Architecture for Production ML at Scale.' MLSys 2019, Stanford, CA.

Technical Blog Posts (Medium / Substack – 42K+ subscribers)

- 'Why Your Kafka Consumer Group Is Lagging (And How to Fix It)' — 180K views, featured in Hacker News #3
- 'Kubernetes Pitfalls: 12 Mistakes That Cost Us \$2M' — 95K views, featured in The New Stack
- 'From Notebook to Production: A Practical MLOps Guide' — 73K views, featured in Towards Data Science
- 'Event Sourcing Isn't Magic: A Realistic Deep-Dive' — 61K views, featured in DZone

CONFERENCE SPEAKING & WORKSHOPS

- KubeCon + CloudNativeCon NA 2023 (Chicago) – 'Scaling Developer Platforms: Lessons from 240 Engineers' [45 min keynote-adjacent]
- AWS re:Invent 2022 (Las Vegas) – 'FinOps Deep Dive: \$4M in Savings and What We Learned' [Workshop, 180 attendees]
- PyCon US 2022 (Salt Lake City) – 'AsyncIO & Kafka in Python: Real-Time Pipelines Without the Pain' [Talk]
- DataEngConf 2021 (Virtual) – 'Feature Stores Demystified: Building vs. Buying for ML Production' [Keynote]
- Kafka Summit 2020 (Virtual) – 'Exactly-Once Semantics in Practice: What Nobody Tells You' [Talk]
- SREcon Americas 2019 (Santa Clara) – 'Error Budgets and SLO Enforcement: A Case Study' [Talk]

OPEN SOURCE CONTRIBUTIONS

- Apache Kafka (apache/kafka) – 17 merged PRs: connector stability improvements, consumer group rebalancing optimizations.
- KafkaLens (github/alexthornton/kafkalens) – Author & maintainer: 4,200+ ⭐, 180+ contributors, 22K+ downloads.
- Backstage (backstage/backstage) – 8 merged PRs: TechDocs plugin improvements, custom scaffolding templates.
- Terraform AWS Provider (hashicorp/terraform-provider-aws) – 4 merged PRs: EKS node group bug fixes.
- KubeFlow (kubeflow/kubeflow) – 6 merged PRs: pipeline caching improvements, notebook controller fixes.

EXTENDED TECHNICAL SKILLS REFERENCE

Programming Languages & Runtimes

- Expert (daily use, 10+ yrs): Python 3.x, Java 8–21, SQL (PostgreSQL, MySQL, Oracle)
- Proficient (frequent use, 5+ yrs): Go 1.18+, TypeScript/JavaScript (ES2020+), Kotlin, Scala, Bash/UNIX Shell
- Working knowledge (project use, 2+ yrs): Rust, C++17, Ruby, R, MATLAB
- AI/ML specific: Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, JAX, Hugging Face Transformers, LangChain, LlamaIndex)

Cloud & Infrastructure (Detailed)

- AWS (14 certs): EC2, EKS, ECS, Lambda, S3, RDS Aurora, DynamoDB, ElastiCache, Kinesis, MSK, SageMaker, Bedrock, CloudFront, Route53, IAM, KMS, Secrets Manager, CodePipeline, CDK, CloudFormation
- GCP: GKE, Cloud Run, BigQuery, Vertex AI, Pub/Sub, Cloud Spanner, Firestore, Cloud Composer (Airflow), Dataflow (Beam), Dataproc (Spark)
- Azure: AKS, Azure Functions, Cosmos DB, Azure DevOps, Azure ML, Event Hubs, Service Bus, Key Vault, Azure AD B2C
- Multi-cloud: Anthos, Crossplane, Pulumi multi-cloud IaC, Cloudflare (DNS, CDN, Workers, Zero Trust)

Data Engineering & Architecture

- Streaming: Apache Kafka, Kafka Streams, ksqlDB, Confluent Platform, Apache Flink, Apache Beam, AWS Kinesis
- Batch: Apache Spark (PySpark, Scala), Databricks, Google Dataproc, AWS Glue, dbt (data build tool)
- Storage: Delta Lake, Apache Iceberg, Apache Hudi, Apache Parquet, Apache ORC, Apache Avro
- Orchestration: Apache Airflow, Dagster, Prefect, AWS Step Functions, Google Cloud Composer
- OLAP: Databricks SQL, Snowflake, Google BigQuery, AWS Redshift, ClickHouse, Apache Druid, Trino/Presto
- Operational DB: PostgreSQL 15, MySQL 8, AWS Aurora, Google Cloud Spanner, CockroachDB, TiDB (NewSQL)
- NoSQL: MongoDB, DynamoDB, Apache Cassandra, Couchbase, Neo4j (Graph), Apache HBase
- Caching & Search: Redis (Cluster + Sentinel), Memcached, Elasticsearch, OpenSearch, Apache Solr

Engineering Methodologies & Practices

- Architecture Patterns: Microservices, Event-Driven Architecture (EDA), CQRS, Event Sourcing, Hexagonal/Clean Architecture, Domain-Driven Design (DDD), Saga Pattern, Strangler Fig
- Reliability Engineering: SLO/SLI/Error Budgets, Chaos Engineering (Gremlin, Chaos Monkey), Blameless Postmortems, Incident Management (PagerDuty, OpsGenie)
- Security: Zero Trust Architecture, OAuth2/OIDC, SAML 2.0, JWT, mTLS, HashiCorp Vault, AWS Secrets Manager, Policy-as-Code (OPA), SAST/DAST (Snyk, SonarQube, Trivy)
- Agile: SAgile (Scaled Agile Framework), Scrum Master certified, Kanban, OKR management, PI Planning facilitation
- API: REST, GraphQL (Apollo Federation), gRPC/Protobuf, AsyncAPI, OpenAPI 3.0, Webhook patterns, API gateway (Kong, AWS API GW, Apigee)

LANGUAGES

English Native / Bilingual C2	Spanish Professional Proficiency B2	Mandarin Chinese Conversational A2
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AWARDS & RECOGNITION

Engineering Excellence Award – Most Impactful Platform Initiative	MegaCorp Financial	2023
Top Performer (Top 2% company-wide)	MegaCorp Financial	2021 & 2022
Tech Innovation Award – Nexus AI Platform	Streamline Analytics Inc.	2019
National Science Foundation Graduate Research Fellowship	NSF	2009–2011
IEEE Student Paper Award – Best Systems Paper	University of Michigan	2009
Hack the North Winner – Best Infrastructure Project	University of Waterloo	2018
Forbes 30 Under 30 – Technology (Nominee)	Forbes Media	2020

IMPACT & METRICS SUMMARY

Quantified Career Impact

\$4.2M/yr Cloud Cost Savings	\$78M Fraud Prevented	2.4B events/day Data Platform Scale
99.999% Uptime SLA	180+ Models ML in Production	240+ Engineers Platform Users
38ms P99 API Latency	14% Churn Reduction	4 hrs Model Deploy Time
4,200+ GitHub Stars (OSS)	17 PRs Apache Kafka Merges	42K+ Blog Subscribers

VOLUNTEER WORK & COMMUNITY

- Code.org Ambassador (2019–Present) – Volunteer instructor for K-12 coding workshops in underserved Bay Area communities; taught 400+ students.
- TechEquity Collaborative (2020–Present) – Board Advisor; support equity initiatives in Bay Area tech hiring and education.
- SF Python Meetup (2018–Present) – Organizer; grew community from 120 to 2,400+ members; speaker coordination for 60+ events.
- Google Summer of Code (GSoC) Mentor (2020, 2021, 2022) – Mentored 5 open-source contributors on distributed systems projects.
- MIT/Harvard Tech Pathways (2021–Present) – Guest lecturer; 'Building for Scale: From 100 to 100M Users' annual lecture.

PROFESSIONAL REFERENCES

Available upon request. The following leaders have agreed to serve as references:

Sarah Chen, VP Engineering MegaCorp Financial Technologies Direct Manager (2020–Present)	Marcus Williams, CTO Streamline Analytics Inc. Direct Manager (2017–2019)
Dr. Jennifer Park, Prof. CS Stanford University M.S. Thesis Advisor (2009–2011)	Kevin O'Brien, Principal Eng. HealthBridge Systems Peer Collaborator (2015–2017)

DECLARATION

I hereby declare that all information provided in this resume is accurate and true to the best of my knowledge. I understand that any false statements may lead to disqualification or termination of employment.

Signature: _____ Date: March 14, 2026

Alexander James Thornton

— End of Resume Document —

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